№3

25 2025

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LaTeX,

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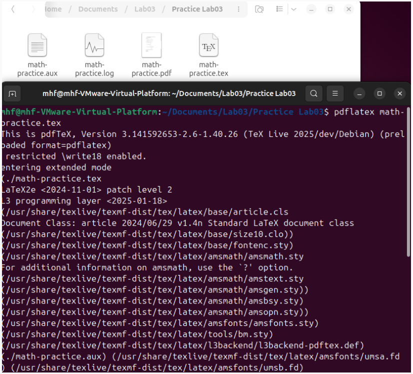
1. LaTeX
2.
3.
4.
5.

: - - : $y = mx + c$ - - : \[y = mx + c \]

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{bm}
\begin{document}

\section{Mathematics Typing - Main Learning}

\subsection{1. Basic Math Modes}
Inline math: $y = mx + c$ \\
Display math:
\[
y = mx + c
\]

\subsection{2. Superscripts and Subscripts}
$x^2$, $a_n$, $x_i^2$, $n_{k-1}^{m+1}$

\subsection{3. Greek Letters and Functions}
$\alpha$, $\beta$, $\theta$, $\Gamma$, $\sin\theta$, $\log x$

\subsection{4. Integration Example}
\[
\int_{-\infty}^{\infty} e^{-x^2} \, dx
\]

\subsection{5. Aligned Equations}
\begin{align*}
a & = b + c \\
x & = y - z \\
```

math-practice.tex :



:

```
pdflatex math-practice.tex
```

-

```
\[
\int_{-\infty}^{\infty} e^{-x^2} \, dx
\]

\subsection{5. Aligned Equations}
\begin{align*}
a & = b + c \\
x & = y - z \\
Q_{n,k} & = Q_{n-1,k} + Q_{n-1,k-1}
\end{align*}

\subsection{6. Numbered Equation and Matrices}
\begin{equation}
e^{i\pi} + 1 = 0
\end{equation}

\[
\begin{pmatrix}
1 & 2 \\
3 & 4
\end{pmatrix}
\]
```

exercises.tex 4 :

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\begin{document}

\section{Exercises from Section 3}

\subsection{Exercise 1: Inline vs Display Math}
Take the same formula and try it both ways: \\
Inline: $f(x) = x^{2} + y^{2}$ \\
Display:
\[
f(x) = x^{2} + y^{2}
\]

\subsection{Exercise 2: Greek Letter Practice}
Try these Greek letters: \\
$\delta$, $\Delta$, $\sigma$, $\Sigma$, $\phi$, $\Phi$
```

:

pdflatex exercises.tex

```
mhf@mhf-VMware-Virtual-Platform:~/Documents/Lab03/Exercises Lab03$ pdflatex exer
cises.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.26 (TeX Live 2025/dev/Debian) (prel
oaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercises.tex
LaTeX2e <2024-11-01> patch level 2
L3 programming layer <2025-01-18>
(/usr/share/texlive/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/06/29 v1.4n Standard LaTeX document class
(/usr/share/texlive/texmf-dist/tex/latex/base/size10.clo))
(/usr/share/texlive/texmf-dist/tex/latex/base/fontenc.sty)
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsmath.sty
For additional information on amsmath, use the `?' option.
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amstext.sty
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsgen.sty))
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsbsy.sty)
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsopn.sty))
(/usr/share/texlive/texmf-dist/tex/latex/amsfonts/amsfonts.sty)
(/usr/share/texlive/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
(./exercises.aux) (/usr/share/texlive/texmf-dist/tex/latex/amsfonts/umsa.fd)
(/usr/share/texlive/texmf-dist/tex/latex/amsfonts/umsb.fd)
Underfull \hbox (badness 10000) in paragraph at lines 22--25
```

. 5:

:

/ : x^2, a_n

: α, β, Γ

: $\sin\theta$, $\log x$

: $\int e^{-x^2} dx$

: pmatrix, bmatrix

: $\mathbf{B}$, $\mathbb{R}$, C

1 Exercises from Section 3

1.1 Exercise 1: Inline vs Display Math

Take the same formula and try it both ways:
Inline: $f(x) = x^2 + y^2$
Display:

:

$\frac{a}{b}$ -

$\sum$, $\prod$ -

$\lim$, $\int$ -

\begin{align*} -

\begin{pmatrix} -

$\mathbf{A}$ -

:

4

1 Mathematics Typing - Main Learning

1.1 1. Basic Math Modes

Inline math: $y = mx + c$
Display math:

$$y = mx + c$$

1.2 2. Superscripts and Subscripts

x^2, a_n, x_i^2, n_{k-1}^{m+1}

1.3 3. Greek Letters and Functions

α, β, θ, Γ, $\sin\theta$, $\log x$

№3:

LaTeX

amsmath

LaTeX .